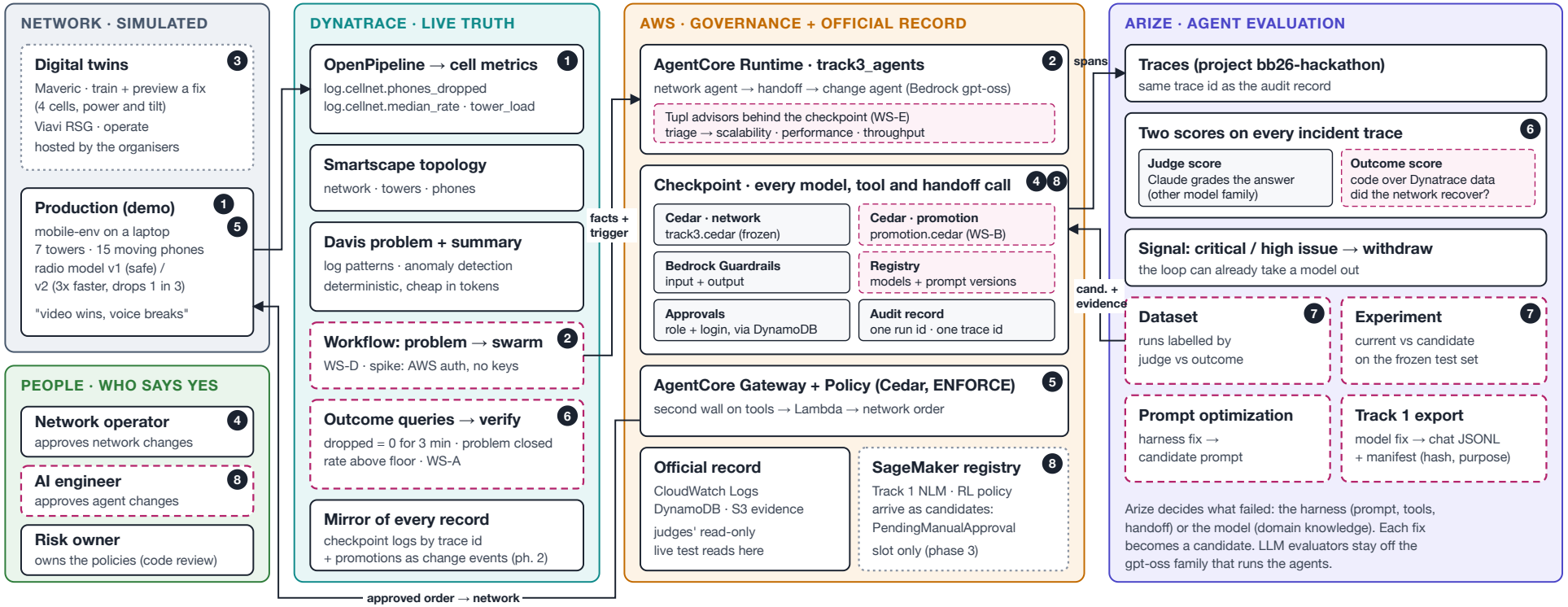
Governed SDLC loop for network agents: target architecture

Dynatrace tells the agents what is true and later confirms whether the fix worked. Arize scores the agents and decides what they learn from. AWS enforces that every change, to a tower or to the swarm itself, is decided by policy, approved by the right person, and recorded.

Breaking Barriers 2026 · team04 · Track 3 primary
draft 2026-11-02 · branch governed-sdlc-loop
built · phase 1 (today) · phase 2–3



THE LOOP · ONE TURN, EVERY STEP WITH A RUN ID

1 Sense	2 Decide	3 Predict	4 Approve	5 Act	6 Verify	7 Learn	8 Promote
BUILT	BUILT	PH. 2	PH. 0	BUILT	PH. 1	PH. 1	PH. 1
Dynatrace problem, metrics, topology	swarm on AgentCore; Workflow trigger (WS-D)	twin preview of the fix before asking	network operator types yes	Gateway + Policy → network order	Dynatrace outcome score (WS-A)	Arize dataset, experiment (WS-C)	AI engineer, through the checkpoint (WS-B)

the promoted agent works the next incident

Network change
e.g. roll back the radio model
Approver: network operator
Enforced: Cedar (track3) + Gateway Policy

Agent change
e.g. new prompt, model or runtime version
Approver: AI engineer
Needs: eval ≥ declared target, no Signal issue

Policy change
e.g. edit a Cedar rule
Approver: risk owner, code review
The loop has no path to it

Same door, different key.
Self-improvement can change how the swarm thinks, never what it is allowed to do. Every change leaves one record.

Numbers on boxes match the loop stages below.

How a change is governed, and how a run is scored

The operator trusts the swarm's network changes. The AI engineer trusts the swarm's own changes. Both go through one checkpoint, under different policies, and both leave the same kind of record.

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Three classes of change

Change	Example	Approver	Evidence in front of the approver	Enforced by
Network	Roll the radio model back to v1	Network operator	Dynatrace problem and cell metrics now; twin preview of the fix (phase 2)	Checkpoint Cedar `track3.cedar` (frozen); AgentCore Gateway Policy, ENFORCE
Agent	New prompt, new model, new runtime version	AI engineer	Experiment ≥ target declared before the run, on the frozen test set; no open critical or high Arize Signal issue	Checkpoint Cedar `promotion.cedar`; registry; then the AgentCore endpoint moves to the new version; SageMaker approval for trained models
Policy	Edit a Cedar rule	Risk owner	Code review	Git and the AgentCore Policy engine. The improvement loop has no path to a policy.

Self-improvement can change how the swarm thinks, never what it is allowed to do. A promoted agent still needs the operator's yes for every network change.

Separation of duties (promotion.cedar)

Request	Decision
AI engineer approves a candidate that met its target, no blocking Signal issue	Allow
An agent tries to promote anything, including itself	Deny
A network operator's approval on an agent change	Deny
A candidate below its declared target	Deny
Promotion while Signal has an open critical or high issue	Deny
Anything in the loop acting on a policy	Deny

Everyone signs in with the same AWS role in the event account, so the approver's role and login are recorded on each approval rather than enforced by IAM. Declared in GAPS.md.

Two scores on every incident

	Judge score	Outcome score
Grades	what the agents said	what happened to the network
By	Claude (Bedrock), another model family than the agents	plain code over Dynatrace data, no model
Coverage-first intent (voice)	1–5 for cause, phones and safety	dropped phones 0 for 3 min within 6 min of the fix; problem closed; median rate above a floor
Throughput-first intent (video)	same	median rate weighted first

When they disagree, the run is labelled

Judge	Outcome	Label	Use
pass	pass	confirmed_good	Track 1 training example
pass	fail	sounded_right_failed	the most valuable hard case
fail	pass	judge_miscalibrated	fix the judge
fail	fail	confirmed_bad	negative example

The official record stays in AWS (CloudWatch, DynamoDB, S3); Dynatrace and Arize mirror it under the same trace id.

Why it matters to operators, and how we get there by noon 3 Nov

A use case is configuration: an intent, the KPIs that score it, the tools agents may use, and the policies that govern them. The loop itself does not change.

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hand-in 12:00 PT 3 Nov · read-only 18:00 PT

Operator challenges this answers

Challenge	Answer
The right data in the operator's and the agent's hands	Dynatrace: problem summary, log patterns, anomaly detection, cell topology; deterministic and cheap in tokens
"Hope for the right change and response"	Predict with the twin before, verify with Dynatrace after
Optimize for app traffic: video-heavy vs real-time voice	The intent sets the outcome score's weights. Radio model v2 is a win for video and an outage for voice.
Impact depends on the business case and technical viability	Intent → outcome score; twin → viability
Autonomous networks need control and governance	One checkpoint, three classes of change, three approvers, one record

What we claim, and what we don't

We claim: the loop is built and exercised end to end, with declared targets, a frozen test set, and graders independent of the agents.

We don't claim: a better Track 1 model (the training step is handed off), or a better real network (every network here is a simulator).

Phases and workstreams

Workstream	Builds	When
Must have: human-approved run + video	A person types yes on an incident run; demo recorded with commit and run id on screen; judging.json updated	by 10:00
WS-A Outcome	`outcome.py`, `verify` run on the incident's trace id, outcome score in Arize	phase 1
WS-B Promotion gate	`promotion.cedar`, prompt registry, `promote` with a live allow and live denies	phase 1
WS-C Arize learn	`dataset`, `experiment` (v1 vs v2 on the frozen state), Track 1 `export`	phase 1
WS-D Dynatrace control plane	Workflow starts the swarm when the problem opens; spike first on AWS auth without access keys	phase 1
WS-E Tupl behind the checkpoint	The Tupl specialists as advisors to the change agent, through the checkpoint	phase 1
Twin preview	What-if on the laptop sim; Maveric once its `.env` arrives	phase 2
AgentCore promotion	Agents read the active prompt; runtime v2; move the endpoint	phase 2
NLM candidate	The organisers' TeleLogs NLM on SageMaker as a second candidate	phase 2
Roadmap	Track 1 fine-tune from the export; RL policy as a governed tool; a second intent on the same loop	phase 3

Built today: the AgentCore runtime, Gateway and Policy, the checkpoint with Cedar, Guardrails, registry and audit; the Claude judge; the Arize Signal gate; Dynatrace sensing; the Dynatrace app. Not built: anything on SageMaker; the Workflow trigger; a run approved by a person.